

Data pipeline and Streamlit application for predicting employee attrition



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Introduction



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Employee Attrition:

Reduction of employees through retirement, resignation, deaths, or layoffs without immediately filling vacancies.

Ruehl, Emma. "Attrition: Definition, Types, Causes & Mitigation Tips." *SHRM*, 15 Apr. 2024, <https://www.shrm.org/topics-tools/news/employee-relations/attrition-definition-types-causes-mitigation-tips>. Accessed 21 July 2026.

Causes of Attrition

- Inadequate compensation and benefits
- Work/life imbalance
- Unrealistic workload expectations
- Ineffective management practices
- Misalignment with workplace culture
- Lack of career advancement opportunities

Cost of Attrition

- Loss of productivity
- Employee burnout
- Loss of company knowledge
- Cost of replacement
- Cost of training

Assemble. "Five Hidden Costs of Employee Attrition." *Forbes*, 21 Mar. 2023, <https://www.forbes.com/sites/forbeseq/2023/03/21/five-hidden-costs-of-employee-attrition/>. Accessed 21 July 2026.

01

Explore HR data

We will explore the HR data set, find trends and correlations with employee attrition.

02

Find and create best predictive model

We will train multiple machine learning models, evaluate their performance, and chose the best model based on our business objectives

03

Create data pipeline and Streamlit application

We create a data pipeline, so we can easily repeat the process using other data sets.

We will also build a Streamlit application to enter employee data and make predictions.

Exploratory Data Analysis

HR datasets from Kaggle:

<https://www.kaggle.com/datasets/davidafolayan/hr-analytics?select=Employee.csv>

Two datasets:

Employee.csv (23 columns and 1470 rows) - descriptive data that includes: Department, Overtime, State, Attrition, etc.

PerformanceRating.csv (11 columns and 6709 rows) - Yearly performance ratings for each employee that includes: Job Satisfaction, Relationship Satisfaction, Self Rating, Manager Rating, etc.

18.5%

Total Attrition Rate

Sales: 24%

HR: 23%

Tech: 16%

Attrition Rate by
department

Data Trends

Top 3 positive correlations with attrition:

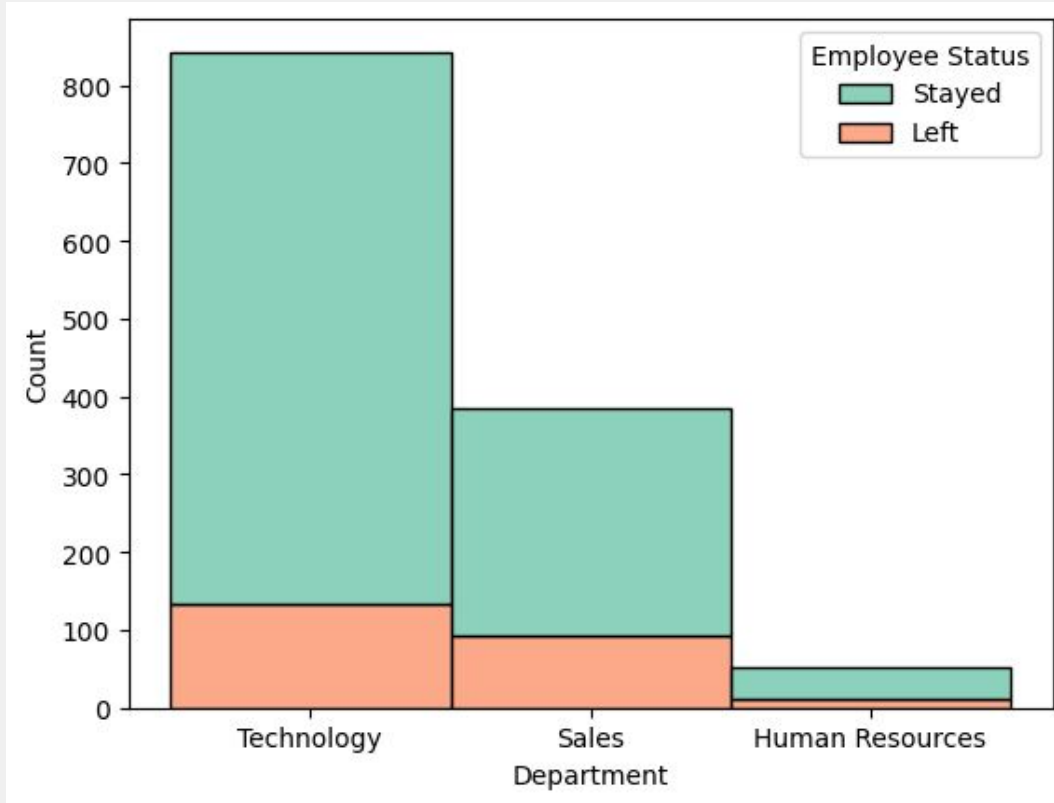
- ◆ Overtime_Yes: 0.26
- ◆ MaritalStatus_Single: 0.20
- ◆ JobRole_Sales Representative: 0.17

Top 3 negative correlations with attrition:

- ◆ YearsAtCompany: -0.43
- ◆ YearsSinceLastPromotion: -0.41
- ◆ YearsinMostRecentRole: -0.30

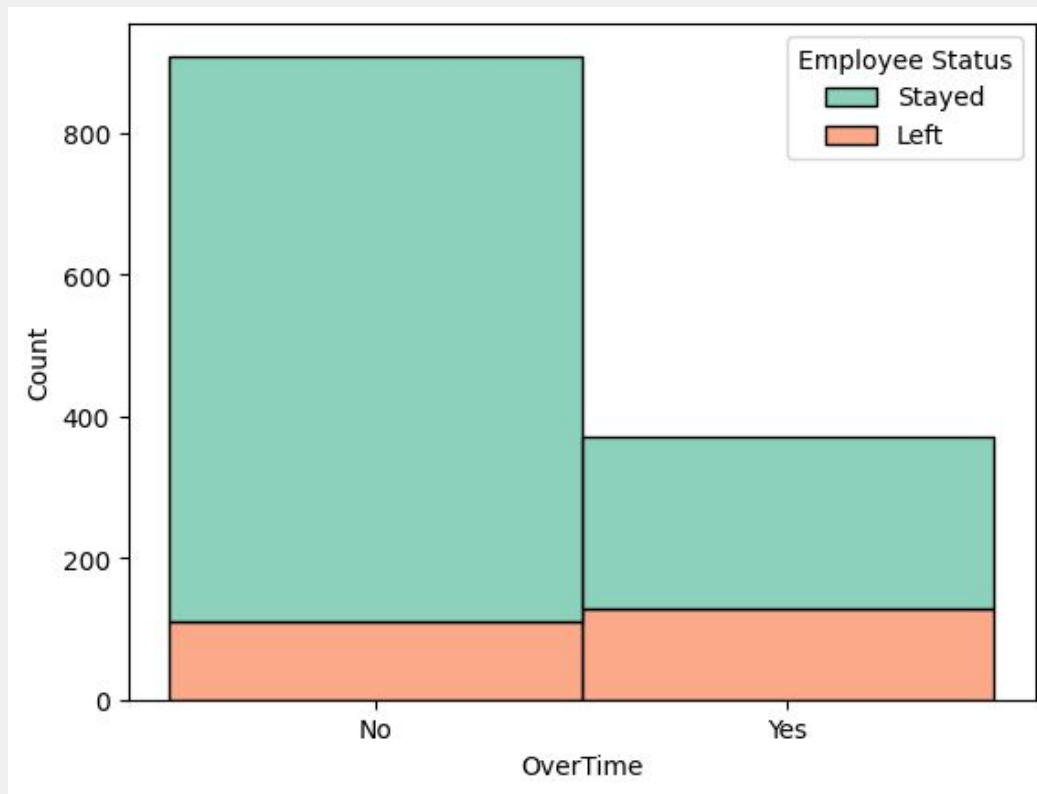
Exploratory Data Analysis

Histogram of Attrition by Department



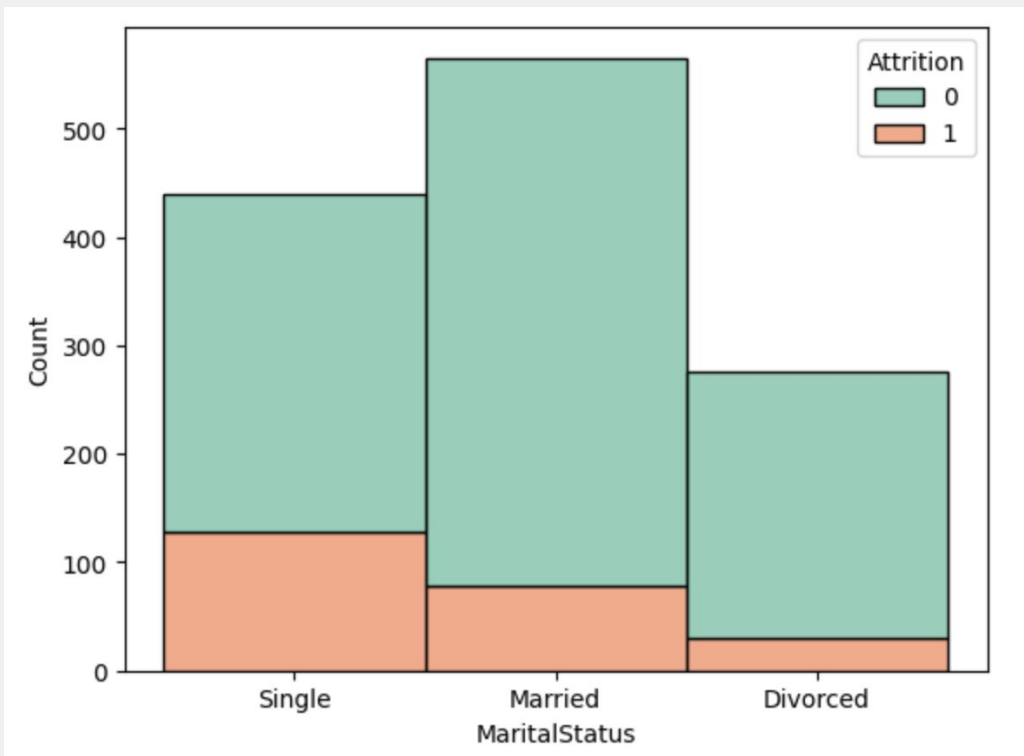
Exploratory Data Analysis

Histogram of Attrition by Overtime



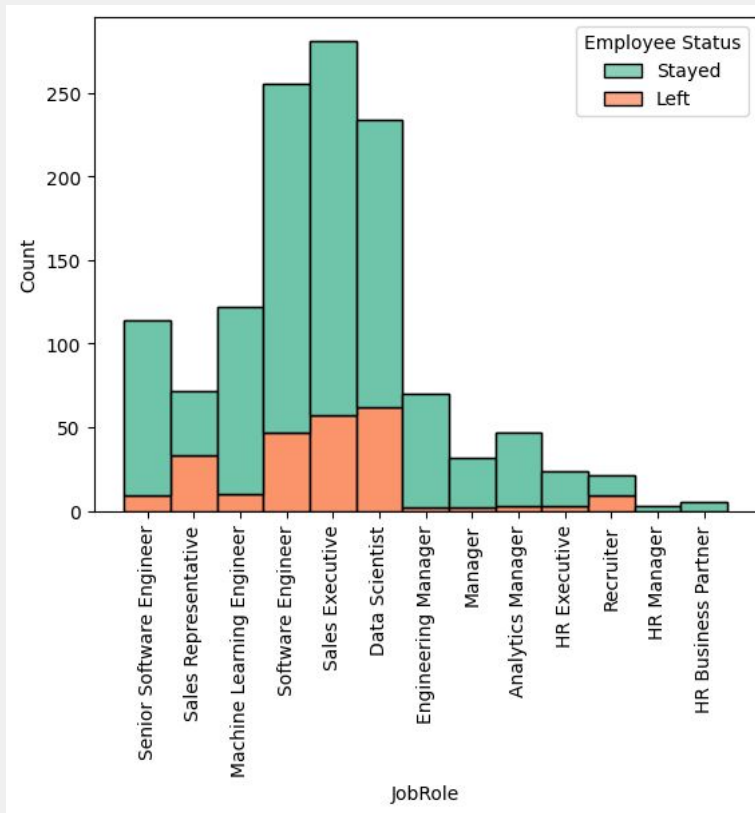
Exploratory Data Analysis

Histogram of Attrition by Marital Status



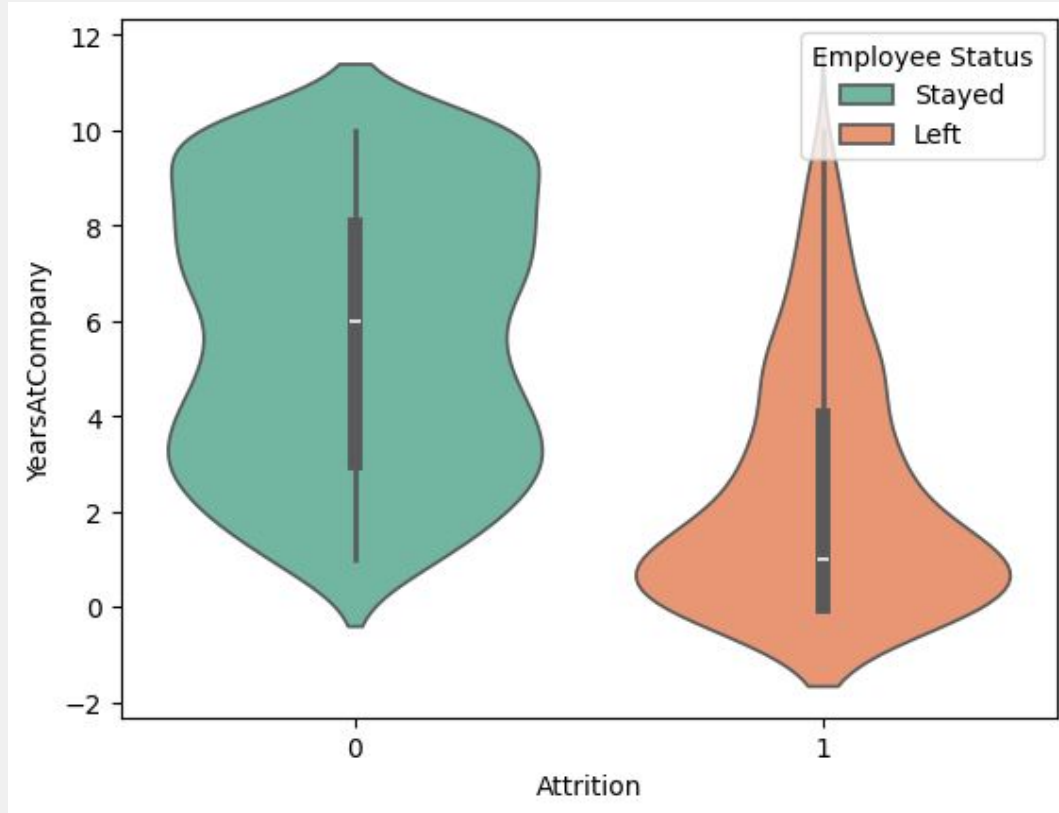
Exploratory Data Analysis

Histogram of Attrition by Job Role



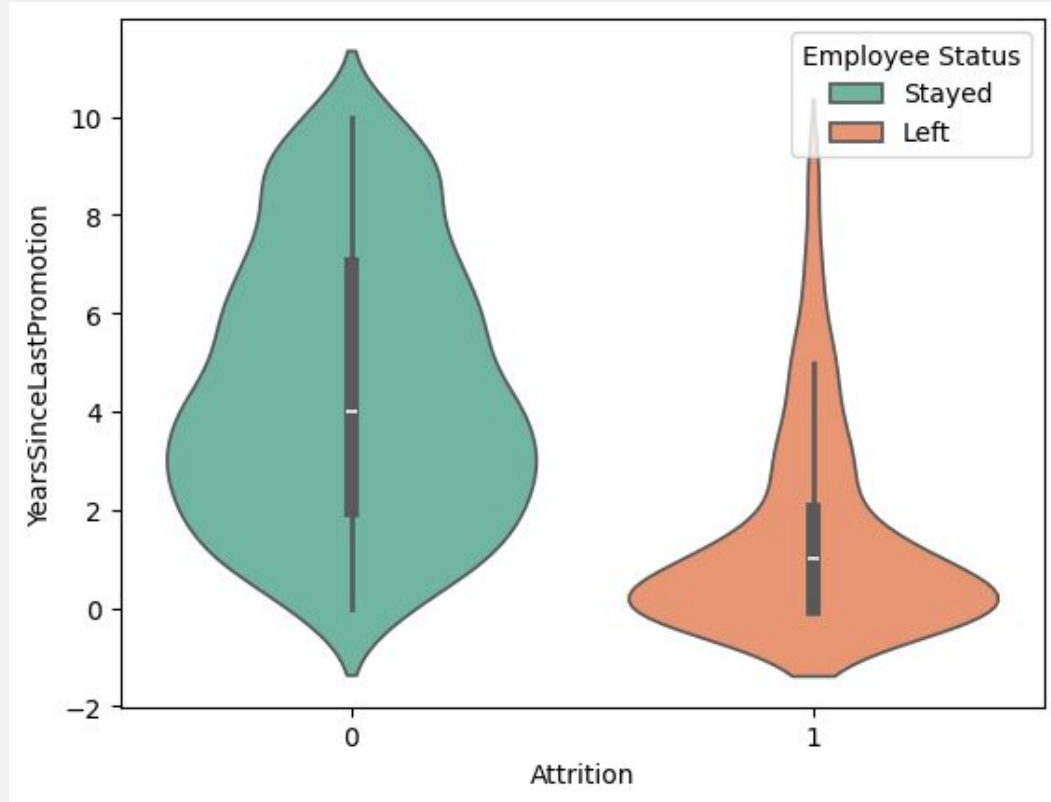
Exploratory Data Analysis

Violin plot of Attrition by Years at Company



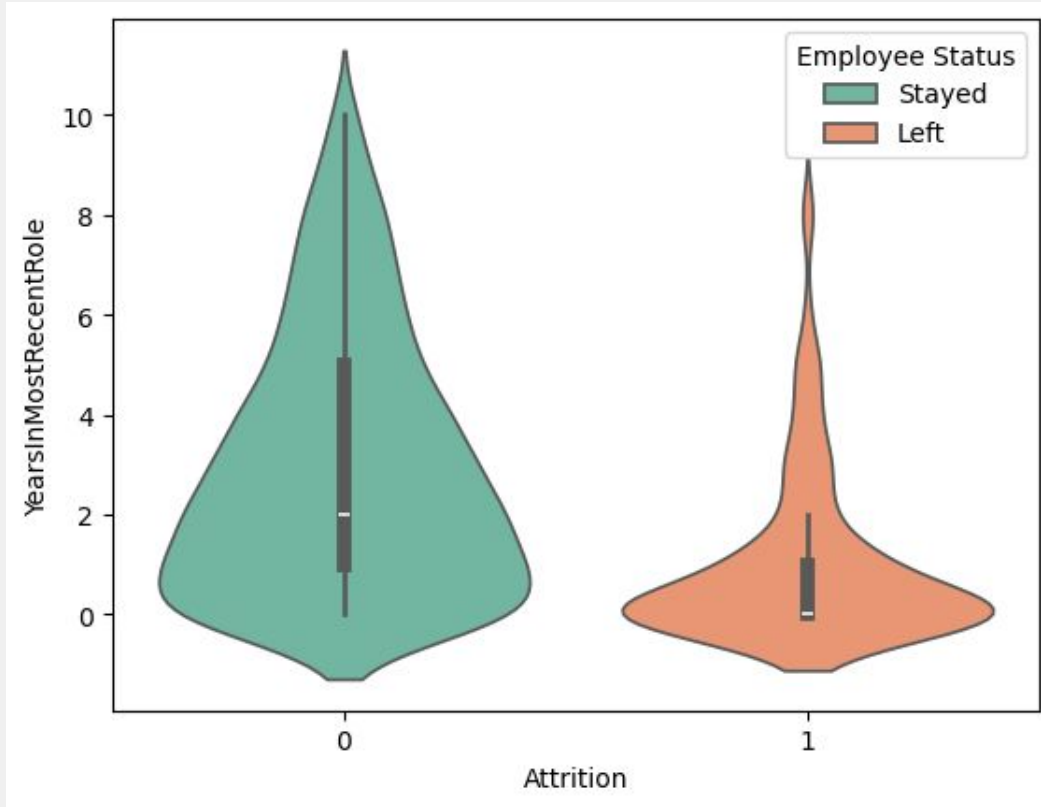
Exploratory Data Analysis

Violin plot of Attrition by Years Since Last Promotion



Exploratory Data Analysis

Violin plot of Attrition by Years in Most Recent Role



Machine Learning (ML) Algorithms

Merge datasets

For performance rating data, we sorted and only kept the latest performance rating for each employee.

Then merged the employee and performance rating data by EmployeeID.

Training and Comparing ML Algorithms

Logistic Regression

Random Forest

Support Vector Machine (SVC)

Adaptive Boosting

K-Nearest Neighbors (KNN)

Gaussian Naive Bayes

Gradient Boosting Classifier

Bootstrap Aggregating)

Machine Learning (ML) Algorithms

Performance Comparison

ML Algorithms	#	Accuracy	#	Precision	#	Recall	#	F1-Score
Logistic Regression		0.898		0.824		0.583		0.683
Support Vector Machine		0.898		0.867		0.542		0.667
Bagging		0.898		0.923		0.500		0.649
AdaBoost		0.891		0.833		0.521		0.641
GBC		0.891		0.833		0.521		0.641
Random Forest		0.895		0.957		0.458		0.620
KNN		0.840		0.578		0.542		0.559
GNB		0.336		0.220		1.000		0.361

Machine Learning (ML) Algorithms

Confusion Matrices

Logistic Regression

True label	Predicted label	
	Stayed	Left
Stayed	202	6
Left	20	28

Random Forest

True label	Predicted label	
	Stayed	Left
Stayed	207	1
Left	26	22

GNB

True label	Predicted label	
	Stayed	Left
Stayed	38	170
Left	0	48

Conclusion & Next Steps

- The Random Forest model has high precision but low recall.
 - Knowing the model has low recall, the business should invest in retention programs to keep employees happy, especially in the first few years
 - Focusing on perform and promotions: mentoring, upskill training, career development planning, etc.
 - Focus on the sales department, especially the sales representatives
- Improving the model with:
 - More data to train the model
 - Feature engineering - did employee's performance decrease from the previous year, did employee's job satisfaction decrease from the previous year, etc.

Data pipeline and Streamlit application

Data cleaning >

Data cleaning function:

Drop unwanted columns

Remove whitespace

Convert yes/no to 1/0

Preprocessor >

Numerical Features
Scaler → imputer &
scaler

Categorical Features →
imputer & encoder

Classifier

Random Forest ML
Algorithm

HR Data Input Features

EnvironmentSatisfaction

5

- +

JobSatisfaction

3

- +

RelationshipSatisfaction

3

- +

TrainingOpportunitiesWithinYear

3

- +

TrainingOpportunitiesTaken

2

- +

WorkLifeBalance

3

- +

SelfRating

2

- +

ManagerRating

2

- +

Gender

Male

▼

Age

Attrition Prediction App

Predict Employee Attrition ↗

Predict

Will employee leave?

Prediction: Yes

Confidence: 57.0%

HR Data Input Features

EnvironmentSatisfaction

5

-

+

JobSatisfaction

4

-

+

RelationshipSatisfaction

5

-

+

TrainingOpportunitiesWithinYear

5

-

+

TrainingOpportunitiesTaken

5

-

+

WorkLifeBalance

5

-

+

SelfRating

5

-

+

ManagerRating

5

-

+

Gender

Male

▼

Age

23

-

+

Attrition Prediction App

Predict Employee Attrition

Predict

Will employee leave?

Prediction: No

Confidence: 96.0%



Any Question?



Thank you!